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## Hydrogen Enrichment of Natural Gas Combustion Using Cantera

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# 1 Introduction

The purpose of this work is to analyse the influence of hydrogen addition on methane combustion. The calculations were performed using the Cantera software.

Methane,  $\text{CH}_4$ , was used as the base fuel. Hydrogen was added to the fuel mixture in different molar fractions. This type of fuel blending is used to investigate the possibility of reducing carbon-related emissions from natural gas combustion.

The main goal of the project was to compare combustion temperature and exhaust gas composition for different hydrogen enrichment levels.

## 2 Hydrogen-Enriched Methane Combustion

With the ongoing decarbonization of the energy and transportation sectors, methane combustion is playing an increasingly important role as a less polluting alternative to conventional fuels. Furthermore, enriching methane with hydrogen can further reduce the environmental impact of the combustion process.

Such fuel mixtures exhibit a wider flammability range, which enables the stable combustion of very lean mixtures. In addition, hydrogen is characterized by a significantly higher flame propagation velocity, which can improve combustion efficiency and reduce the formation of particulate matter in the exhaust gases.

The combustion of such mixtures in piston engines is a particularly interesting research topic. Engines powered by compressed natural gas (CNG) still represent a relatively small niche, especially in road transport and even more so in aviation. Nevertheless, their use in marine propulsion systems and public transportation vehicles is becoming increasingly common and important. In the future, these technologies may also find applications in low-emission propulsion systems for aviation.

The aim of this work is to analyse the influence of hydrogen addition to methane on combustion temperature and combustion product composition. In particular, the effect of hydrogen enrichment on the emissions of  $\text{CO}_2$ ,  $\text{CO}$  and nitrogen oxides will be evaluated using equilibrium calculations performed in the Cantera software.

## 3 Analysed Cases

The analysed fuel mixtures were:

- 100%  $\text{CH}_4$ ,
- 95%  $\text{CH}_4$  + 5%  $\text{H}_2$ ,
- 90%  $\text{CH}_4$  + 10%  $\text{H}_2$ ,
- 80%  $\text{CH}_4$  + 20%  $\text{H}_2$ ,
- 60%  $\text{CH}_4$  + 40%  $\text{H}_2$ .

Air was modelled as:

$$\text{air} = \text{O}_2 + 3.76\text{N}_2 \tag{1}$$

The initial conditions were:

$$T_0 = 700 \text{ K} \quad (2)$$

$$p = 20 \text{ atm} \quad (3)$$

The air excess ratio was varied in the range:

$$\lambda = 0.7 - 1.5 \quad (4)$$

The equivalence ratio used in Cantera was calculated as:

$$\phi = \frac{1}{\lambda} \quad (5)$$

## 4 Calculations

The calculations were carried out in Python using the Cantera library. Mixtures were described using the **gri30.yaml** chemical combustion mechanism from Cantera. It includes species such as methane, hydrogen, oxygen, nitrogen and the main combustion products such as CO, CO<sub>2</sub>, etc.

We defined different air excess ratios, fuel compositions (levels of hydrogen enrichment), and oxidizer composition, and let Cantera calculate the mixture proportions using the **gas.set\_equivalence\_ratio()** function. The equilibrium state after an adiabatic constant-pressure combustion process was then determined using the **gas.equilibrate("HP")** function.

The analysed exhaust gas components were:

- CO<sub>2</sub>,
- CO,
- H<sub>2</sub>O,
- O<sub>2</sub>,
- NO,
- NO<sub>2</sub>,
- unburned H<sub>2</sub>.

## 5 Results

Below are presented the results obtained with our python program.

## 5.1 Adiabatic Flame Temperature

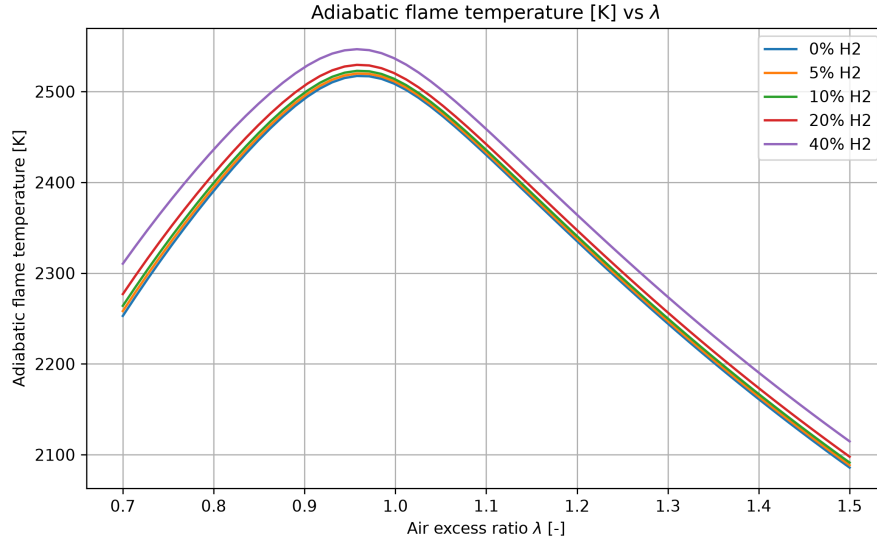


Figure 1: Adiabatic flame temperature as a function of air excess ratio

The maximum combustion temperature occurs in regions close to the stoichiometric mixture. For rich mixtures, the temperature decreases due to incomplete combustion. For lean mixtures, the excess air absorbs part of the released heat, which also decreases the final temperature. There is also rise in flame temperature with an increase in the hydrogen content, however it's not significant, since flame temperatures of hydrogen and methane don't differ much. [1]

## 5.2 Carbon Dioxide Emissions

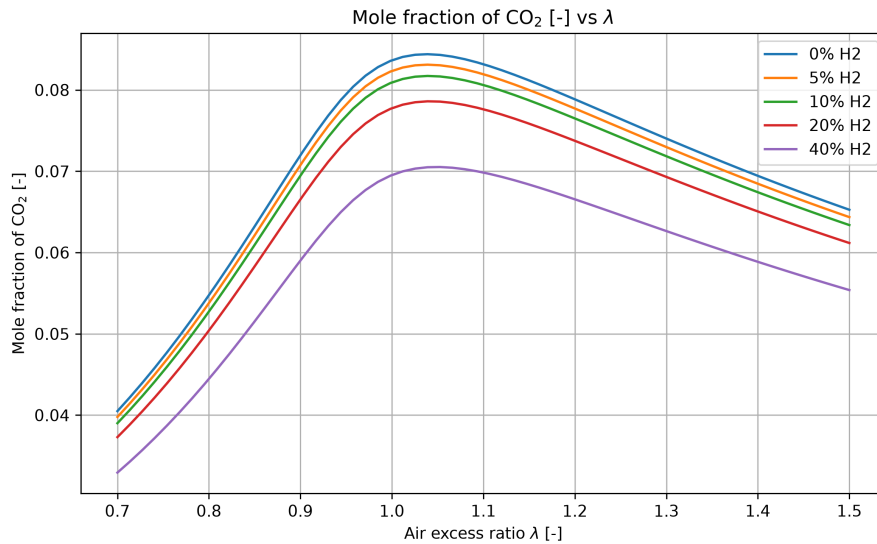


Figure 2: Molar fraction of CO<sub>2</sub> in exhaust gases

Hydrogen addition reduces the amount of carbon in the fuel mixture. Therefore, higher hydrogen enrichment generally decreases the molar fraction of CO<sub>2</sub> in the exhaust gases.

What is also worth pointing out is that the combustion of lean mixtures rapidly decreases the amount of produced carbon dioxide, and the use of these lean mixtures is permitted by an increase in the hydrogen content. Therefore, using enriched fuel decreases the amount of produced  $\text{CO}_2$  in two ways.

### 5.3 Carbon Monoxide Emissions

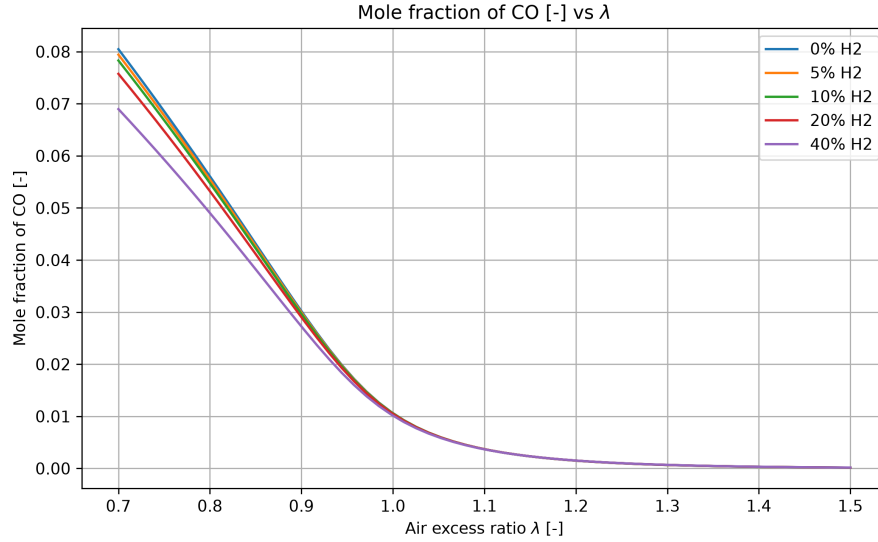


Figure 3: Molar fraction of CO in exhaust gases

Carbon monoxide concentration is highest for rich mixtures. This is caused by insufficient oxygen available for complete oxidation of carbon to  $\text{CO}_2$ . Once lean-mixture conditions are reached, the amount of produced CO is minimal, and it can still be oxidized during exhaust.

In both carbon monoxide and dioxide the simple rule is that the less methane is burned the less emission is produced, and that can be achieved by addition of hydrogen to mixtures.

## 5.4 Water Vapour

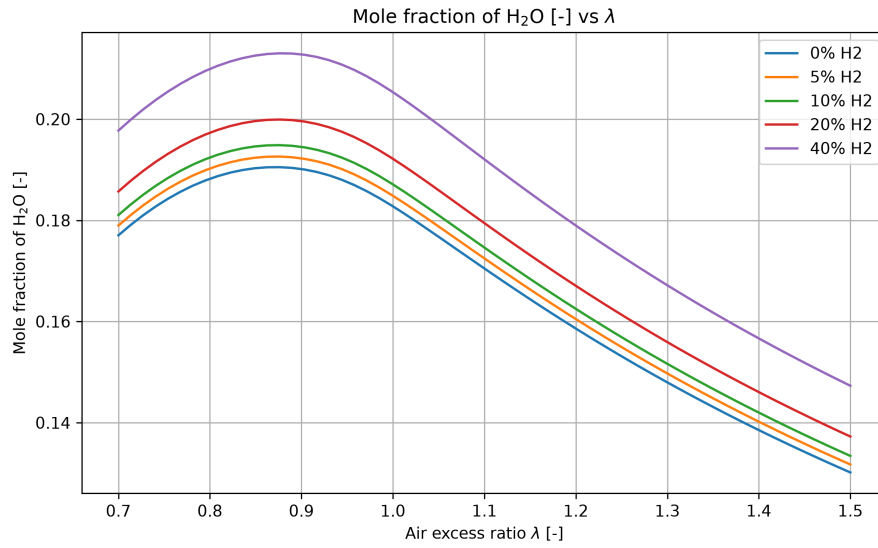


Figure 4: Molar fraction of H<sub>2</sub>O in exhaust gases

The addition of hydrogen increases the amount of hydrogen in the fuel mixture. As a result, water vapour becomes an important combustion product. Even though emission of H<sub>2</sub>O is not harmful, it is important to keep that in mind during design and exploitation of an engine, as it can decrease flame temperature.

## 5.5 Oxygen Concentration

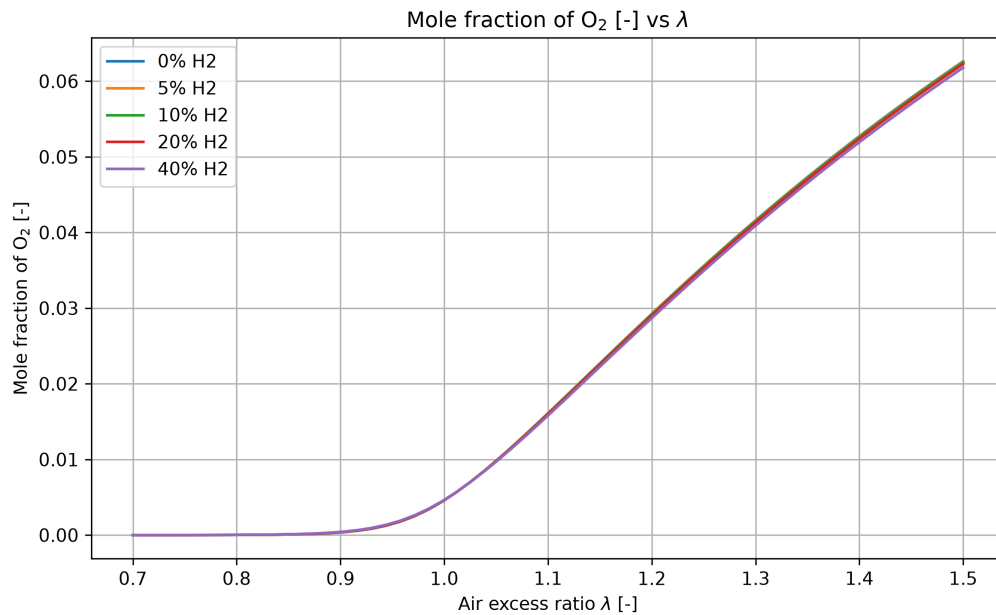


Figure 5: Molar fraction of O<sub>2</sub> in exhaust gases

For lean mixtures, excess oxygen remains in the exhaust gases. For rich mixtures, oxygen is almost completely consumed. As for correlation with hydrogen content next to no difference can be observed.

## 5.6 Nitric Oxides

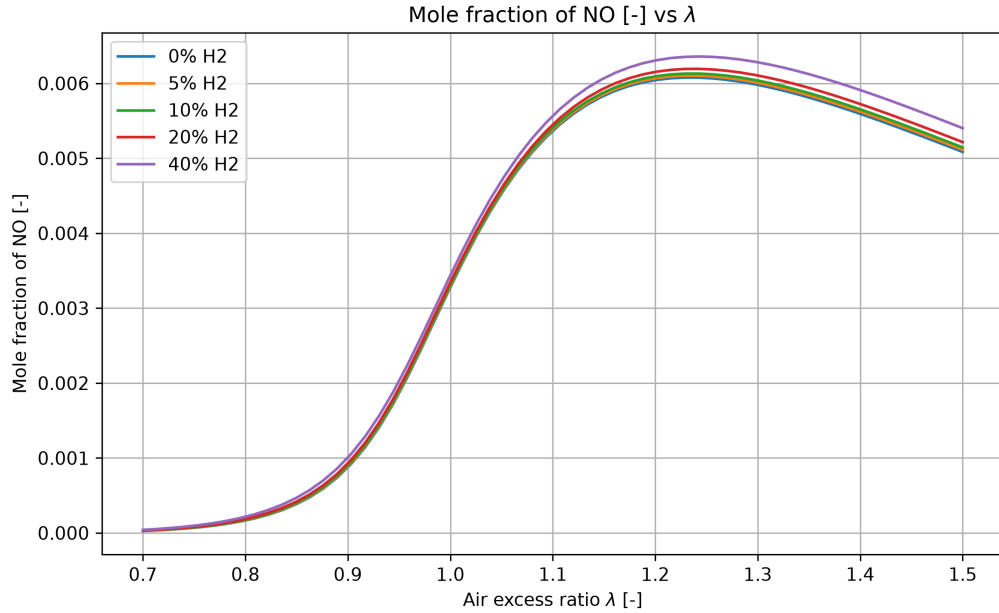


Figure 6: Molar fraction of NO in exhaust gases

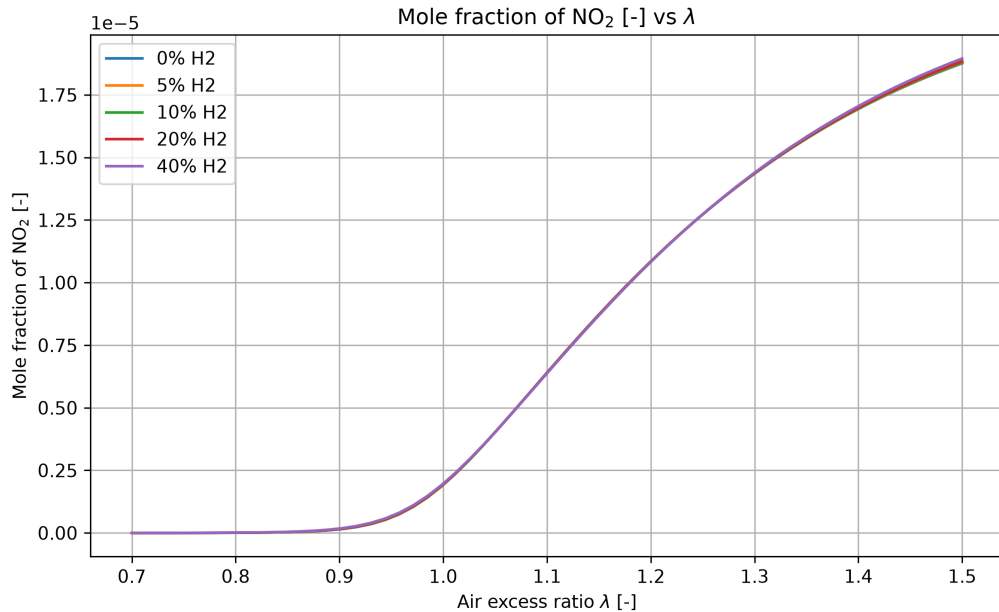


Figure 7: Molar fraction of NO<sub>2</sub> in exhaust gases

NO formation is strongly related to combustion temperature. The highest NO concentration is expected near the region of maximum flame temperature. Very rich and very lean



mixtures lead to lower temperatures and therefore lower NO formation.

## 5.7 Unburned Hydrogen

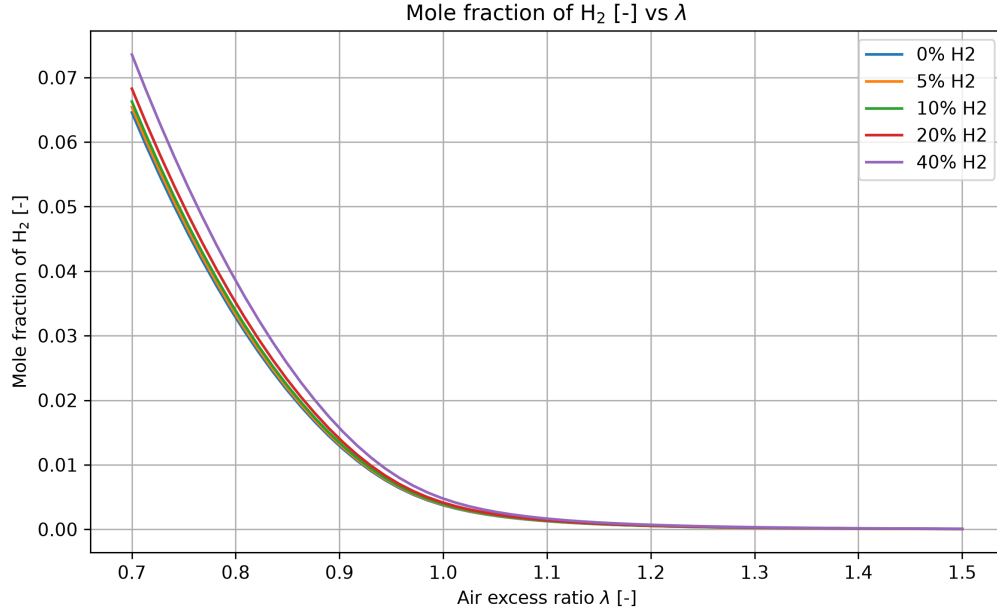


Figure 8: Molar fraction of unburned  $H_2$  in exhaust gases

For rich mixtures and high hydrogen enrichment levels, part of the hydrogen may remain in the exhaust gases.

## 6 Results

The results show that enriched methane with hydrogen affect the combustion process. It comes as no surprise that reducing amount of carbon in mixture reduces amount of carbon-oxygen formations in exhaust. That is especially true with carbon dioxide, as carbon monoxide depends mainly on the air excess ratio.

Rich mixtures promote CO formation due to oxygen deficiency. Lean mixtures reduce CO but increase oxygen concentration in exhaust gases.

Hydrogen addition can also affect NO formation. If the combustion temperature increases, NO concentration may increase as well. However, for very lean or very rich mixtures, the temperature decreases and NO formation is limited.

## 7 Conclusions

The analysis led to the following conclusions:

- Hydrogen enrichment reduces the carbon content of the fuel mixture.
- Increasing hydrogen fraction generally reduces  $CO_2$  concentration in exhaust gases.
- Rich mixtures increase CO concentration due to incomplete oxidation.

- Lean mixtures increase  $O_2$  concentration in the exhaust gases.
- NO formation is strongly connected with combustion temperature.
- High hydrogen addition may increase the amount of unburned  $H_2$  in rich mixtures.
- Hydrogen enrichment changes both combustion temperature and exhaust gas composition
- As for the engine application, hydrogen enrichment seems to be a promising alternative to ordinary natural gases

The most important conclusion is that hydrogen addition can reduce carbon-related emissions from methane combustion, but it may also influence NO formation and the amount of unburned hydrogen in exhaust gases.

## References

- [1] Osama A. Marzouk, Corlell University. *Adiabatic Flame Temperatures for Oxy-Methane, Oxy-Hydrogen, Air-Methane, and Air-Hydrogen Stoichiometric Combustion using the NASA CEARUN Tool, GRI-Mech 3.0 Reaction Mechanism, and Cantera Python Package*. 2025. URL: <https://arxiv.org/abs/2503.11826> (visited on 06/13/2026).